Adaptive Practice Platform: Overview

The adaptive practice platform helps learners improve at programming by adjusting the difficulty of each coding question. The system records correctness, execution time and the number of attempts for every question a learner answers. These performance signals are collected during practice sessions and are used to estimate the learner's current level.

Method

A Random Forest classifier is trained on the performance signals. Its prediction selects beginner, intermediate or advanced practice for the next question. The classifier was chosen because it handles small tabular datasets well and gives feature importances that can be inspected. Training used a synthetic dataset generated for this demonstration.

Limitations

This is a demonstration document. The synthetic dataset is small and does not represent real learners, so the results should not be generalised. The platform does not yet account for learners who guess, and it has not been evaluated against a human tutor.